

Effect of particle size on wear behaviour of SiC particulate-reinforced aluminum alloy composites

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Particle size is one of the most important factors in determining wear properties of particulate-reinforced metal composites. However, it appears to be difficult to draw a fundamental conclusion from the reports about this problem. Some reports have suggested that wear resistance of the composites increased with increasing particle size [1, 2], while others indicated that an increase in particle size had a negligible influence on the wear rate [3]. A further problem is that nearly all the studies have been carried out with such methods as pin-on-disc or sand rubber wheel abrasion tests, in which the sliding speed was maintained in a narrow range and the applied load in a steady state. In fact, among those potential applications of the composites, many cases are characterized by both sliding and impact wear with sliding speed varying over a large range [4]. It is thus necessary to study the effect of particle size on wear properties of the composites under a variety of experimental conditions. In this work, the effect of particle size on wear behaviour of SiC particulate-reinforced 2024Al composites has been investigated using three tests: sliding wear, impact abrasion and erosion.

Composites containing 15 vol % SiC particles of 3.5, 10 and 20 μm in size were prepared by the P/M process and machined into specimens $30 \times 25 \times 3$ mm. The sliding wear test (SWT) was performed on a reciprocating grooving tester. The slider was made of Si₃N₄ ceramic with the top plane 2×2 mm. The sliding speed was 0.01 m s^{-1} and the normal load about 2.2 MPa. The impact abrasion test (IAT) was conducted on a single pendulum scratch tester. It could simulate a single event of abrasive scratching with impact across a surface to produce an arc-like groove and allow measurement of the energy dissipated in the event. The stylus on the end of the pendulum hammer was made of a hard alloy. The initial swinging angle was set to 60° and entrance scratching speed was about 2.1 m s^{-1} . Details of the test are given in paper [5]. The erosion test (ET) was carried out on a sand blasting erosion

tester. 100 mesh corundum was used as erodent, impact speed was about 23 m s^{-1} and the impingement angle was 90° .

The results of the three tests are shown in Table I. For different tests, different measures were adopted to evaluate the wear rate or wear resistance.

For the SWT test, the volume loss of the composites decreases rapidly with increasing particle size. The worn surface of composite 1 is characterized by much plastic deformation and obvious evidence of ploughing or cutting (Fig. 1). Although some cutting tracks can be found on the surface of composite 3, the plastic deformation is obviously less than that of composite 1, and the surface particles form many protrusions and present some evidence of machining by the hard ceramic slider (Fig. 2).

For the IAT test, the specific energy e , defined as the scratching energy for unit volume loss, can be used for evaluating the wear resistance. The higher e , the better the wear resistance. Table I shows the specific energy of the specimens corresponding to a groove volume of $15 \times 10^{-3} \text{ mm}^3$. It can be seen that the wear resistance of the composites decreases slightly with increasing particle size, presenting an adverse tendency to that in the sliding wear test. The

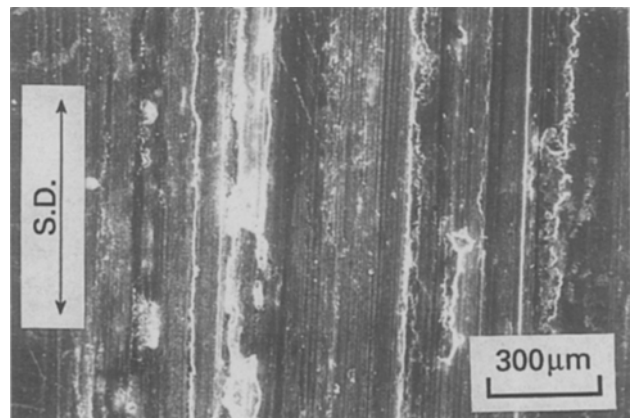


Figure 1 SEM micrograph of a worn surface of composite 1 in the SWT test.

TABLE I Results of the tests

Composites		Hardness HB (kg/mm ²)	SWT Volume loss ($\times 10^{-3} \text{ mm}^3$)	IAT e (J/mm ³)	ET Weight loss (mg)
Number	Particle size (μm)				
1	3.5	89.1	14.3	1.43	2.58
2	10	88.4	8.7	1.28	3.02
3	20	87.2	7.0	1.21	3.56

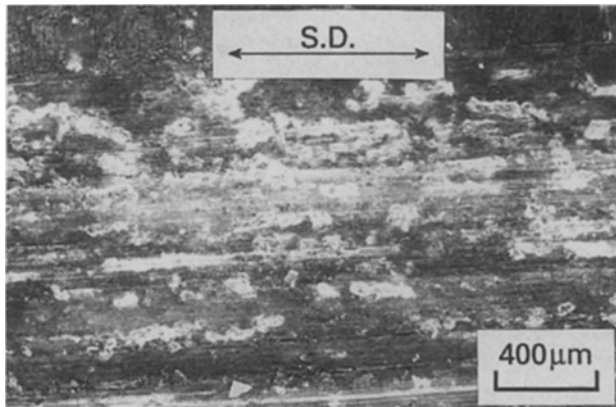


Figure 2 SEM micrograph of a worn surface of composite 3 in the SWT test.

groove surface of composite 1 is smooth, and the debris is stripe-like and contains many shear bands, indicating relatively greater capacity of plastic deformation of the composite during scratching (Fig. 3). For composite 3 the groove surface is characterized by obvious evidence of disbonding or pull-out of the surface particles and fractures of the surface layer (Fig. 4).

For the ET test, weight loss of the composites increases relatively rapidly with increasing particle size. The worn surfaces are covered by impact craters, humps and ridges due to repeated impacting (Fig. 5).

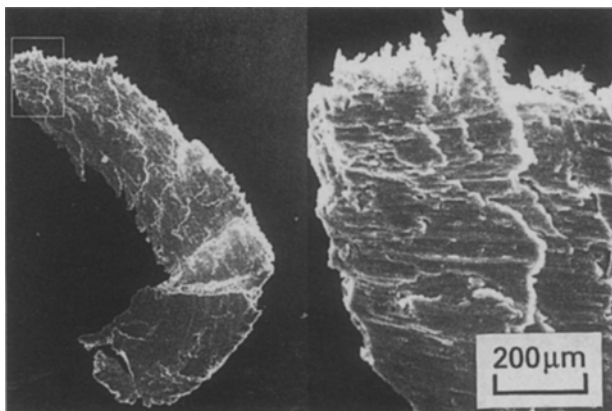


Figure 3 SEM micrograph of a debris of composite 1 in the IAT test.

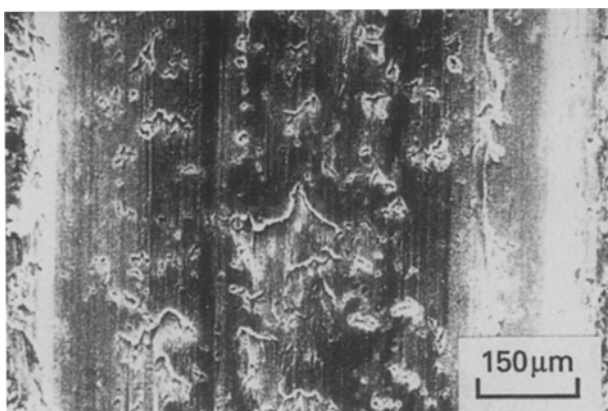


Figure 4 SEM micrograph of a groove surface of composite 3 in the IAT test. The scratching direction is from up to down.

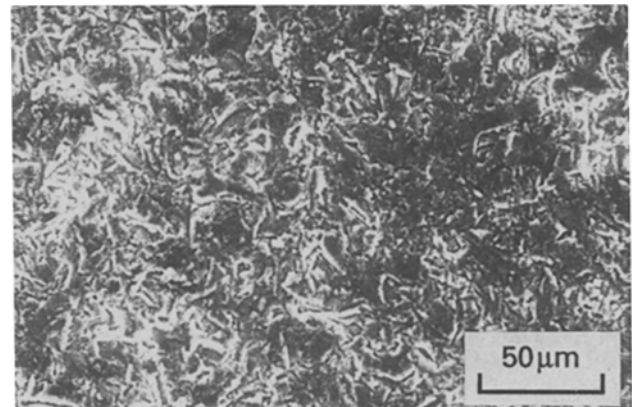


Figure 5 SEM micrograph of a worn surface of composite 3 in the ET test.

The results of the tests are summarized in Fig. 6. The wear behaviour of the composites is clearly dependent on the test conditions. The impact component in both load and sliding speed increase in the order SWT, IAT and ET. In the SWT test, the wear resistance of the composites increases rapidly with increasing particle size. With the impact component in load increasing, the wear resistance present an adverse tendency.

In the SWT test, there was almost no impact in the load and the hard ceramic slider could cut or plough both the matrix and SiC particles. The main wear mechanism was cutting or ploughing and hardness played a dominant role as described in Rabinowicz' law [6]. For composite 1, although the bulk hardness was the greatest among those of the tested composites (Table I), the surface particles could be relatively easily removed along with debris during sliding due to their small size. The slider could penetrate and cut deeply into the surface (Fig. 1), resulting in much material loss. With increasing particle size, surface particles were not so easily removed. During the sliding process, surface particles were maintained in the matrix and formed many protrusions to bear the majority of the load. In fact,

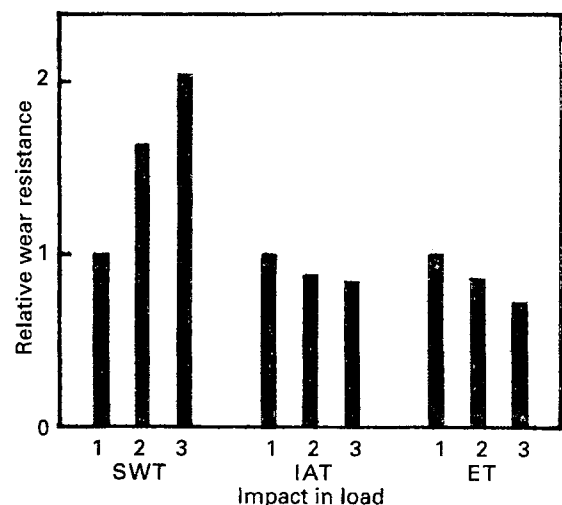


Figure 6 A comparison of wear resistance of the composites in the three tests. The composites are numbered in the order given in Table I. The wear resistance values have been normalized to that of composite 1.

the wear process of composite 3 was basically a process of the slider machining the surface particles (Fig. 2), and the surface hardness of the composite was determined mainly by that of the hard SiC particles. The wear resistance was therefore obviously better than that of the composite containing small particles. It is thus clear that the anti-wear mechanisms of the composites are different for different particle size in sliding wear test. The wear resistance was dependent on the real surface hardness rather than the bulk hardness.

In the IAT test, the wear process was characterized by a combination of cutting and impacting, and a careful balance should be considered between hardness and toughness. For composite 1, on the one hand, the matrix was strengthened effectively by the Orowan strengthening-mechanism [7] due to small interparticle space; on the other hand, the small particles, like the precipitated phases in an alloy hardened by precipitation, could coordinate with the matrix in deformation and allow the composite to present a relatively large capacity for plastic deformation during the scratching process (Fig. 3), which suggests good toughness of the composite. With increasing particle size, the bulk hardness or strength of the composites decrease because the interparticle space increases rapidly and the Orowan mechanism gradually fails to function. The large interparticle space also reduces restraint among the particles and allows the surface particles to be relatively easily broken or pulled out (Fig. 4). What is worse is that large particles cannot coordinate with the matrix during deformation. During a scratching process, the surface particles act as large defects, producing many fractures in the surface or subsurface layer, thus reducing the surface toughness. Impact abrasive resistance therefore decreases with increasing particle size.

In the ET test, the impingement angle was 90° and the wear mechanism was dominated by deformation wear (Fig. 5). The wear behaviour was therefore mainly dependent on the toughness of the materials

rather than hardness. The SiC particles were hard but brittle, and were likely to be broken and removed under high-speed impacting. The larger the particles were, the more likely they were to be broken or removed. Moreover, composites containing small particles possessed better plasticity than those containing larger particles, as stated above. So the erosion resistance of the composites containing small particles is obviously better than that of the composites containing larger particles.

From the discussion above, it can be seen that the wear behaviour of particulate-reinforced aluminum composites is significantly affected by the particle size. Composites containing large particles exhibit excellent wear resistance under sliding wear conditions with a steady applied load. Under wear conditions characterized by some impact component in the load, composites containing small particles are preferred.

Acknowledgements

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